Prime-Leaf Tree Theory

Arithmetic as the statistical mechanics of multiplication histories

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Abstract

We treat the integers as a gas. The single-particle spectrum is the set of *primons*: one mode per prime p, of energy $\log p$. In the usual reading (the primon gas of Julia and of Bost–Connes), an integer is a Fock state (its prime factorization is an occupation-number configuration) and the Riemann zeta function is the free bosonic partition function. This paper studies the same gas *before* symmetrization and *before* the interaction history is forgotten. A microstate is then a *multiplication tree*: a fireball built by recorded pairwise fusions of primons, ordered and parenthesized, whose only conserved quantity is the integer $N(T) = e^{E(T)}$ it evaluates to. Multiplication is not commutative or associative at this level, because microstates remember their formation; ordinary arithmetic is the fully coarse-grained description, and the Fundamental Theorem of Arithmetic is precisely the statement that the occupation-number description is faithful (zero residual entropy). Forgetting history and symmetrizing are independent coarse-grainings, so there are four gases, each with an exact partition function determined by the prime zeta function $P(\beta)$: the Riemann gas $\zeta(\beta)$, a distinguishable-particle gas with inverse Hagedorn temperature $1.39943\ldots$, a non-planar fireball gas at $1.98847\ldots$, and the full planar fireball gas, whose partition function obeys the statistical bootstrap equation $Z = P + Z^2$ of Hagedorn–Frautschi type and reaches its Hagedorn point at $\beta_H = \sigma^* = 2.59738\ldots$, where $P(\sigma^*) = \frac{1}{4}$. There the partition function is finite and exactly $\frac{1}{2}$, with the square-root branch point universal for branched polymers: the transition is continuous, whereas the symmetrized gases diverge at theirs. Associativity is, in this precise sense, a first-order axiom. The complex-β singularities of the fireball gas (its Fisher zeros) are computed: they inherit the Riemann zeros through P, and their new branch points, the solutions of $P(\beta) = \frac{1}{4}$, do *not* lie on a vertical line. Finally, the bridge to addition: fireballs act as *processes* on states of hierarchically clustered matter. Under this action associativity becomes unobservable (processes compose), commutativity fails at the level of states and survives only in expectation, and distributivity is a theorem about the extensive observable rather than an axiom. Primes themselves emerge as the cluster sizes that cannot crystallize into identical cells. All combinatorial claims are verified by exhaustive enumeration and all constants to high precision (Appendix A).

Contents

1 Introduction

1.1 Multiplication before evaluation

Ordinary arithmetic writes

$$3 \times 2 = 6 \qquad \text{and} \qquad 1+1+1+1+1+1 = 6$$

and treats the two results as the same kind of object. The starting intuition of this paper is that they are not. Six liters obtained by pouring is a flat additive quantity. “Three groups of two” is a *structured* object, different from "two groups of three", and its measured value happens to be 6. The slogan:

> *You never get six liters from multiplication. You get a structured object whose shadow is* 6.

Physics has a standard name for this situation: 6 is a *macrostate*, and the ways of building it are *microstates*. We therefore refuse to let multiplication collapse into numbers. A product is a *multiplication tree*: the recorded fusion history of the primes involved, ordered and parenthesized, unevaluated. The familiar integer is produced only by an evaluator, the shadow map N, which plays the role of the measurement: it reads off the single conserved quantity and discards the history. The classical laws of multiplication are then not properties of multiplication at all. They are properties of the *coarse-graining*: statements about which distinct microstates the measurement fails to distinguish. In one line:

> *The primes are the particles; an integer is a calorimeter reading; the Riemann zeta function is a partition function; and ordinary arithmetic is what survives total coarse-graining.*

1.2 The dictionary

The paper is written for readers who think in statistical mechanics. Every physical term below is either an *exact structural identity* (the object on the left satisfies, verbatim, the defining equation of the object on the right) or is flagged as imagery. Theorems remain theorems; the physics is load-bearing, not decorative.

tree theory	physics
prime p	primon: single-particle mode of energy $\varepsilon_p = \log p$
multiplication tree $T \in \mathcal{T}$	fireball: a recorded history of pairwise fusions
$\boxtimes$ (pair formation)	fusion; on energies $E = \log N$ it is addition
shadow $N(T)$	the calorimeter reading: the measured integer $e^{E(T)}$
left–right order kept	ordered channels, as for planar diagrams
parenthesization kept	interaction history retained
quotient by associativity	forget the history, keep the beam: asymptotic state
quotient by commutativity	Bose symmetrization
words $\mathcal{W}$	basis of the full (Boltzmann) Fock space
multisets $\mathcal{M}$	occupation numbers: the primon gas of Julia
degeneracy $g(n)$; $S(n) = \log g(n)$	microstate count and Boltzmann entropy of the integer n
Fund. Thm. of Arithmetic	the occupation-number description is faithful: $S_{\mathcal{M}} \equiv 0$
$Z(s) = \sum N^{-s}$, $s = \beta$	canonical partition function at inverse temperature β
$Z = P + Z^2$	statistical bootstrap equation (Hagedorn–Frautschi–Nahm)
critical exponents σ	inverse Hagedorn temperatures β_H
solutions of $P(s) = \frac{1}{4}$	Fisher singularities: complex-β singularities of Z
grouped quantities $\mathcal{G}$	states of hierarchically clustered matter
the action $\rho(T)$	a process applied to states; $\boxtimes \mapsto$ composition
flattening α	the extensive observable (particle number); expectation level

Table 1: The dictionary. "Calorimeter", "fireball", "beam" and "cluster" are imagery; every other row is an exact identity established in the text.

Everything is elementary and self-contained: finite combinatorics plus one-variable complex analysis. No result depends on the imagery.

1.3 Results

1. **The microstate space** (§2). The fireball world is the free magma $\mathcal{T}$ on the set of primes; the measurement is the unique magma morphism $N\colon \mathcal{T} \to (\mathbb{N}_{\geq 2}, \cdot)$ with $N(L_p) = p$; energies $E = \log N$ add under fusion. The interacting world has no vacuum: 1 is not the shadow of any fireball, and the empty state exists only in the Fock description.

2. **Coarse-graining is two-dimensional** (§3). Forgetting the interaction history (associativity) and symmetrizing (commutativity) are *independent* operations, giving a commuting square of four state spaces $\mathcal{T}, \mathcal{W}, \mathcal{B}, \mathcal{M}$. The Fundamental Theorem of Arithmetic says exactly that the final corner has one microstate per macrostate. Each level carries a computable entropy: e.g. $S_{\mathcal{T}}(12) = \log 6$. The degeneracies are Catalan $\times$ multinomial for $\mathcal{T}$, multinomial for $\mathcal{W}$ (OEIS A008480), a sequence for $\mathcal{B}$ that restricts to the Wedderburn–Etherington numbers on prime powers and appears not to have been recorded before, and 1 for $\mathcal{M}$. All verified exhaustively for $n \leq 20000$.

3. **Four partition functions and a Hagedorn ladder** (§4). All four are exact functionals of the prime zeta function P: the free Bose gas $\zeta(\beta)$ (the Euler product *is* the textbook free-boson computation), a geometric gas $P/(1-P)$, an Otter gas with argument-doubling equation, and the planar fireball gas obeying the bootstrap equation $Z = P + Z^2$. The inverse Hagedorn temperatures form a strict ladder

$$1 \;<\; 1.3994333287263303 \;<\; 1.9884768518009081 \;<\; \sigma^* = 2.5973851271346716,$$

   one rung per layer of remembered structure. At its Hagedorn point the fireball gas converges with $Z_{\mathcal{T}}(\sigma^*) = \frac{1}{2}$ exactly and a square-root branch point: the universal branched-polymer singularity, the same criticality that governs planar-diagram counting in the Gaussian matrix model. The symmetrized gases instead have poles: whether the Hagedorn temperature is attainable is decided by associativity. The classical Hagedorn–Frautschi–Nahm bootstrap, transplanted onto the primon spectrum, slots into the same family with $\beta_H = 2.1487511659\ldots$ (Boltzmann counting) and $2.3072314551\ldots$ (Bose counting).

4. **Fisher zeros** (§5). $Z_{\mathcal{T}}$ is algebraic in P, so its complex-β singularities are inherited from P (which carries a branch point at $\beta = \rho/k$ for every nontrivial Riemann zero ρ, and has the imaginary axis as a natural boundary), plus new branch points where $P(\beta) = \frac{1}{4}$. We compute the five with $0 < \operatorname{Im}\beta < 50$: they recur near integer multiples of $2\pi/\log 2$ (the lightest primon sets the frequency) and do *not* lie on a vertical line. The naive tree analogue of the Riemann Hypothesis is false; the right question is the distribution of these Fisher singularities, and that question we answer: in every substrip of $\operatorname{Re} s > 1$ the tree zeros have an exact asymptotic density (Theorem 5.3, via the Jessen–Tornehave theory of almost periodic functions). Each singularity occurs when a single primon mode outshines the coherent rest, and the density is the total dominance rate $\frac{1}{2\pi}\sum_p (\log p)\, q_p$. A census of all 59 zeros to height 500 matches the formula's Monte Carlo prediction of 58.3 to one percent.

5. **The additive bridge** (§6). Fireballs act as *processes* on states of clustered matter (boxed heaps of unit quanta), and the classical laws stratify by observability: associativity is invisible to the action (exactly, because processes compose), while commutativity fails on states and holds only in expectation, and distributivity holds only in expectation. "You never get six liters from multiplication" becomes a theorem about the image of the action. The primes themselves are then derivable inside the additive world: a size is composite iff its heap can crystallize into $k \geq 2$ identical cells of $m \geq 2$ quanta; primes are the non-crystallizable sizes.

6. **Computational content** (§8). Two theorems mark what the gas can and cannot compute. Microstate counting is *shape-blind*: every degeneracy depends only on the exponent profile of n, so no counting of histories can locate primes: the no-free-lunch face of freeness. In the other direction, a random integer seen at full resolution is a universal random object: a uniform Catalan tree with a Gaussian, $\log\log x$, number of primons (Erdős–Kac lifted to fireballs). The interface where the theory genuinely meets the complexity of finding primes is *expression cost*: integer complexity is the first invariant here that escapes P (it distinguishes 4 from 9), Wilson's theorem makes primality one factorial-expression away, and the open frontier, deterministic prime generation, is a derandomization problem, mapped but not shortened.

1.4 Relation to existing work

The Fock-level reading is classical: the primon gas of Julia [22] and Spector [34], deepened by Bost–Connes [5], identifies $\zeta(\beta)$ with a free bosonic partition function over the spectrum $\{\log p\}$ and locates a Hagedorn catastrophe at $\beta = 1$. Hagedorn's limiting temperature and the statistical bootstrap are from [20, 18], with the analytical solution and the square-root criticality due to Nahm [29]. Lee–Yang and Fisher zeros are from [26, 15]. On the mathematical side, the free magma and its Catalan combinatorics are standard [35]; the degeneracy $g_{\mathcal{T}}(n)$ appears as OEIS A144757 and the ordered-factorization count as A008480 [33]; ordered factorizations with arbitrary parts go back to Kalmár [24]; the natural boundary of the prime zeta function is due to Landau and Walfisz [25]. Closest in spirit to our refusal to symmetrize is Loday's *arithmetree* [27, 6]: arithmetic on planar binary trees with non-commutative addition and one-sided distributivity.

Against this background the contributions here are: the two-dimensional coarse-graining (the square, with the new non-planar corner, its Otter equation and its Hagedorn point); the exact critical value $Z_{\mathcal{T}}(\sigma^*) = \frac{1}{2}$ and the attainability dichotomy; the computation and non-collinearity of the Fisher zeros; the process/state/expectation stratification of the arithmetic laws; and the internal derivation of the primes. One caution runs through the paper: a *free* structure carries no secrets about the primes themselves. Everything classically hard enters through the single-particle spectrum, that is through $P(\beta)$, or through the additive bridge. What the theory offers is a clean separation of the structural from the arithmetic, and computable invariants in the separated layers.

2 The fireball algebra

2.1 Primons and fusion histories

Definition 2.1 (Multiplication trees)**.** Fix the set of prime leaves $\{L_p : p \text{ prime}\}$. The set $\mathcal{T}$ of *multiplication trees* is defined by the grammar

$$T \ ::= \ L_p \ | \ (T_1 \boxtimes T_2).$$

Thus $\mathcal{T}$ is the *free magma* on the set of primes: $\boxtimes$ is ordered-pair formation, subject to no laws whatsoever.

Physical reading. L_p is a single primon, of energy $\varepsilon_p = \log p$. A tree is a fireball that remembers how it was assembled: $T_1 \boxtimes T_2$ is the fusion of the fireballs T_1 and T_2, in that order. Examples of *distinct* microstates:

$$L_2, \qquad L_2 \boxtimes L_3, \qquad L_3 \boxtimes L_2, \qquad (L_2 \boxtimes L_3) \boxtimes L_5, \qquad L_2 \boxtimes (L_3 \boxtimes L_5).$$

Non-commutativity and non-associativity are not axioms we impose; they are the *absence* of axioms. Keeping the left–right order is keeping the fusion channels ordered, as one does for planar diagrams of matrix-valued fields [37]; keeping the parenthesization is keeping the interaction history: which pair fused first. A multiplication tree is the parse tree of a multiplication expression before evaluation; equivalently, a tree-level diagram built from a single cubic vertex whose external legs are primons.

For $T \in \mathcal{T}$ write $|T|$ for the number of leaves and $\mathrm{word}(T)$ for the left-to-right sequence of leaf labels; $\mathrm{supp}(T)$ denotes the set of primes occurring.

2.2 The measurement

Definition 2.2 (Shadow)**.** The *shadow map* $N : \mathcal{T} \to \mathbb{N}_{\geq 2}$ is defined recursively by

$$N(L_p) = p, \qquad N(T_1 \boxtimes T_2) = N(T_1)\, N(T_2).$$

Equivalently: N is the unique magma morphism from $\mathcal{T}$ to the multiplicative semigroup $(\mathbb{N}_{\geq 2}, \cdot)$ sending L_p to p.

Physical reading. Define the energy $E(T) = \log N(T)$. Then $E(L_p) = \varepsilon_p$ and $E(T_1 \boxtimes T_2) = E(T_1) + E(T_2)$: energies add under fusion *because* shadows multiply. The shadow is the calorimeter reading: the single conserved quantity, blind to history and to order. An integer is not a microstate; it is the measured value e^E shared by many microstates.

Remark 2.3 (No vacuum)*.* $\mathcal{T}$ has no unit object and 1 is not the shadow of any tree: the image of N is exactly $\mathbb{N}_{\geq 2}$. One could freely adjoin an empty tree I, but then $I \boxtimes T \neq T$ (freeness again), which buys nothing. In gas language: the interacting world has no vacuum state; the vacuum ($n = 1$, zero energy) exists only in the Fock description, as the empty occupation configuration (§3).

Composite numbers are not primitive objects of this world. There is no L_6; the value 6 occurs only as the common reading of the two fireballs $L_2 \boxtimes L_3$ and $L_3 \boxtimes L_2$. Likewise 8 is the common reading of $(L_2 \boxtimes L_2) \boxtimes L_2$ and $L_2 \boxtimes (L_2 \boxtimes L_2)$: distinct formation histories, one observable. Quantifying this degeneracy is the subject of the next section.

3 Coarse-graining: from histories to Fock space

3.1 Two independent coarse-grainings

Definition 3.1. Let $\approx_a$ be the smallest congruence on $(\mathcal{T}, \boxtimes)$ with $(T_1 \boxtimes T_2) \boxtimes T_3 \approx_a T_1 \boxtimes (T_2 \boxtimes T_3)$ for all T_i, and $\approx_c$ the smallest congruence with $T_1 \boxtimes T_2 \approx_c T_2 \boxtimes T_1$. Write

$$\mathcal{W} = \mathcal{T}/{\approx_a}, \qquad \mathcal{B} = \mathcal{T}/{\approx_c}, \qquad \mathcal{M} = \mathcal{T}/({\approx_a} \vee {\approx_c}).$$

Each quotient is a familiar free object: $\mathcal{W}$ is the free semigroup on the primes (nonempty *words* in primes, i.e. ordered prime factorizations); $\mathcal{B}$ is the free *commutative* magma (trees with unordered branches); $\mathcal{M}$ is the free commutative semigroup (nonempty finite *multisets* of primes). The shadow map descends to all four corners, and the two identifications commute:

$$\begin{tikzcd}
\mathcal{T} \arrow[r, "\text{forget history}"] \arrow[d, "\text{symmetrize}"'] & \mathcal{W} \arrow[d, "\text{symmetrize}"] \\
\mathcal{B} \arrow[r, "\text{forget history}"'] & \mathcal{M}
\end{tikzcd}$$

The shadow map of the final corner is where classical arithmetic begins:

Proposition 3.2 (FTA as exactness of the collapse)**.** *The induced map $N\colon \mathcal{M} \to \mathbb{N}_{\geq 2}$ is a bijection. This statement is equivalent to the Fundamental Theorem of Arithmetic.*

Proof. Elements of $\mathcal{M}$ are nonempty prime multisets $\{p_1, \ldots, p_k\}$ with $N = p_1 \cdots p_k$. Surjectivity of N is the existence of a prime factorization; injectivity is its uniqueness. $\square$

Physical reading. The two arrows are the two things an observer can throw away. *Forgetting the history* keeps only the ordered list of constituent primons: the asymptotic content of the fireball, an S-matrix state with the internal dynamics integrated out. *Symmetrizing* declares the primons indistinguishable: Bose statistics. Doing both lands in the occupation-number description: an element of $\mathcal{M}$ is exactly a configuration $(\alpha_2, \alpha_3, \alpha_5, \ldots)$, a basis vector of the symmetric Fock space over the single-particle spectrum. $\mathcal{W}$, the half-way house that keeps order but forgets history, is a basis of the *full* (Boltzmann) Fock space. Proposition 3.2 then says: *the occupation-number description of the integers is faithful.* Unique factorization is the absence of residual entropy at the Fock level.

Remark 3.3 (The identifications are themselves structured). The associativity identification is not a group action: two trees with the same word are connected by a sequence of reassociations, and the set of parenthesizations of a fixed word carries the *Tamari lattice* structure, realized geometrically by the associahedron [36]. Commutativity moves similarly generate the permutohedron. The quotient square therefore has polytopal fibers, a refinement we do not pursue here.

3.2 Degeneracies: the entropy of an integer

For $n \geq 2$ with factorization $n = \prod_p p^{\alpha_p}$ write $\Omega(n) = \sum_p \alpha_p$. For each corner $\mathcal{X}$ of the square let

$$g_{\mathcal{X}}(n) = \#\{X \in \mathcal{X} : N(X) = n\}, \qquad S_{\mathcal{X}}(n) = \log g_{\mathcal{X}}(n),$$

the microstate count and the Boltzmann entropy of the macrostate n at level $\mathcal{X}$. Proposition 3.2 says $S_{\mathcal{M}} \equiv 0$; the other three levels have genuine entropy.

Proposition 3.4 (Planar fireballs)**.**

$$g_{\mathcal{T}}(n) \;=\; C_{\Omega(n)-1}\, \frac{\Omega(n)!}{\prod_p \alpha_p!}, \qquad C_k = \frac{1}{k+1}\binom{2k}{k}.$$

Proof. A tree with shadow n has exactly $\Omega(n)$ leaves, by induction and multiplicativity of N. The map $T \mapsto (\mathrm{shape}(T), \mathrm{word}(T))$ is a bijection from $\{T : N(T) = n\}$ to (binary tree shapes with $k = \Omega(n)$ leaves) $\times$ (sequences of primes with product n). There are C_{k-1} shapes [35] and $\Omega!/\prod \alpha_p!$ sequences (permutations of the prime multiset). The two factors are independent: the fiber of the full collapse $\mathcal{T} \to \mathcal{M}$ splits as histories $\times$ orderings. □

Proposition 3.5 (Words)**.** *$g_{\mathcal{W}}(n) = \Omega(n)!/\prod_p \alpha_p!$, with recursion $g_{\mathcal{W}}(n) = \sum_{p|n} g_{\mathcal{W}}(n/p)$ and $g_{\mathcal{W}}(p) = 1$. This is OEIS A008480.*

Proposition 3.6 (Non-planar fireballs)**.** *$g_{\mathcal{B}}$ satisfies $g_{\mathcal{B}}(p) = 1$ and, for composite n,*

$$g_{\mathcal{B}}(n) \;=\; \frac{1}{2} \sum_{\substack{d|n \\ 1<d<n}} g_{\mathcal{B}}(d)\, g_{\mathcal{B}}(n/d) \;+\; \frac{1}{2}\,[\, n \textit{ is a square}\,]\; g_{\mathcal{B}}(\sqrt{n}).$$

On prime powers, $g_{\mathcal{B}}(p^k)$ is the k-th Wedderburn–Etherington number (OEIS A001190): 1, 1, 1, 2, 3, 6, 11, 23, ...

Proof. An element of $\mathcal{B}$ over composite n is an unordered pair $\{B_1, B_2\}$ with $N(B_1)N(B_2) = n$. Ordered pairs are counted by $\sum_{1<d<n} g_{\mathcal{B}}(d) g_{\mathcal{B}}(n/d)$; the diagonal (possible only when n is a square, both factors of shadow $\sqrt{n}$ and equal as trees) is counted by $g_{\mathcal{B}}(\sqrt{n})$; unordered pairs with repetition are (ordered + diagonal)/2. On p^k all leaves are identical and the count reduces to unordered binary trees with k indistinguishable leaves, the defining recursion of the Wedderburn–Etherington numbers. □

Remark 3.7 (A sequence not in the OEIS). At the time of writing, the sequence $g_{\mathcal{B}}(2), g_{\mathcal{B}}(3), \cdots = 1, 1, 1, 1, 1, 1, 1, 1, 1, 1, 2, \ldots$ does not appear in the OEIS. The closest entry, A375120 ("complete binary unordered tree-factorizations"), uses a different convention: it counts the two children as an *ordered* pair whenever the factor split is (d, d), so it disagrees with Proposition 3.6 first at $n = 144$ (it gives 42 where the true unordered count is 41) and disagrees with Wedderburn–Etherington on prime powers. Submitting $g_{\mathcal{B}}$ is a small item of independent interest.

Example 3.8 (The four levels over $n = 12$)**.**

$\mathcal{T}:$	6 trees	$(L_2 \boxtimes L_2) \boxtimes L_3,\ (L_2 \boxtimes L_3) \boxtimes L_2,\ (L_3 \boxtimes L_2) \boxtimes L_2,$ $L_2 \boxtimes (L_2 \boxtimes L_3),\ L_2 \boxtimes (L_3 \boxtimes L_2),\ L_3 \boxtimes (L_2 \boxtimes L_2);$
$\mathcal{W}:$	3 words	$(2,2,3),\ (2,3,2),\ (3,2,2);$
$\mathcal{B}:$	2 bushes	$\{\{L_2, L_2\}, L_3\},\ \{\{L_2, L_3\}, L_2\};$
$\mathcal{M}:$	1 multiset	$\{2,2,3\}.$

So the entropy of 12 is $S_{\mathcal{T}}(12) = \log 6$ at the history level, $S_{\mathcal{W}}(12) = \log 3$ for the beam, $S_{\mathcal{B}}(12) = \log 2$ non-planar, and 0 in Fock space. And $g_{\mathcal{T}}(12) = C_2 \cdot \frac{3!}{2!\,1!} = 2 \cdot 3 = 6$, as predicted.

Table 2 lists the three nontrivial counts for small n. All formulas and recursions of this subsection were verified against brute-force enumeration for every $n \leq 20000$ (Appendix A).

n	Ω	$g_{\mathcal{T}}$	$g_{\mathcal{B}}$	$g_{\mathcal{W}}$	n	Ω	$g_{\mathcal{T}}$	$g_{\mathcal{B}}$	$g_{\mathcal{W}}$
2	1	1	1	1	20	3	6	2	3
4	2	1	1	1	24	4	20	4	4
6	2	2	1	2	27	3	2	1	1
8	3	2	1	1	28	3	6	2	3
10	2	2	1	2	30	3	12	3	6
12	3	6	2	3	32	5	14	3	1
16	4	5	2	1	36	4	30	6	6
18	3	6	2	3	64	6	42	6	1

Table 2: Degeneracies over n at the three structured levels (primes and other rows with all counts 1 omitted). Note $g_{\mathcal{T}} = C_{\Omega-1} \cdot g_{\mathcal{W}}$ throughout, and $g_{\mathcal{B}}(2^k) = 1, 1, 1, 2, 3, 6$ for $k = 1, \ldots, 6$ (Wedderburn–Etherington).

4 Partition functions: bootstrap equations and the Hagedorn ladder

4.1 Four gases, four functional equations

Write $E(X) = \log N(X)$ and let $\beta = s$ be the inverse temperature. Let $P(s) = \sum_p p^{-s} = \sum_p e^{-s\varepsilon_p}$ denote the prime zeta function, physically the single-particle (single-primon) partition function. For each corner $\mathcal{X} \in \{\mathcal{T}, \mathcal{W}, \mathcal{B}, \mathcal{M}\}$ define the canonical partition function

$$Z_{\mathcal{X}}(s) \;=\; \sum_{X \in \mathcal{X}} e^{-sE(X)} \;=\; \sum_{X \in \mathcal{X}} \frac{1}{N(X)^s} \;=\; \sum_{n \geq 2} \frac{g_{\mathcal{X}}(n)}{n^s}$$

(for $\mathcal{M}$ we include the empty configuration, of shadow 1, so that $Z_{\mathcal{M}} = \zeta$ exactly).

Theorem 4.1 (Functional equations)**.** *As identities of Dirichlet series,*

$$Z_{\mathcal{T}}(s) = P(s) + Z_{\mathcal{T}}(s)^2, \qquad Z_{\mathcal{T}} = \frac{1 - \sqrt{1 - 4P(s)}}{2} = \sum_{k \geq 1} C_{k-1}\, P(s)^k, \tag{4.1}$$

$$Z_{\mathcal{W}}(s) = P(s) + P(s)\, Z_{\mathcal{W}}(s), \qquad Z_{\mathcal{W}} = \frac{P(s)}{1 - P(s)} = \sum_{k \geq 1} P(s)^k, \tag{4.2}$$

$$Z_{\mathcal{B}}(s) = P(s) + \tfrac{1}{2}\big(Z_{\mathcal{B}}(s)^2 + Z_{\mathcal{B}}(2s)\big), \qquad Z_{\mathcal{B}} = 1 - \sqrt{1 - 2P(s) - Z_{\mathcal{B}}(2s)}, \tag{4.3}$$

$$Z_{\mathcal{M}}(s) = \zeta(s) = \prod_p \frac{1}{1 - p^{-s}}, \qquad \zeta = \exp\Big(\sum_{k \geq 1} \frac{P(ks)}{k}\Big). \tag{4.4}$$

In (4.1) *and* (4.3) *the branch of the square root is fixed by $Z \to 0$ as $\operatorname{Re} s \to \infty$.*

Proof. (4.1): a tree is a leaf or an ordered pair of trees, and N^{-s} is multiplicative over $\boxtimes$, so the sum over pairs factors as $Z_{\mathcal{T}}^2$. Solving the quadratic and expanding gives the Catalan series. (4.2): a word is a prime or a prime followed by a word. (4.3): an element of $\mathcal{B}$ is a leaf or an unordered pair $\{B_1, B_2\}$, possibly equal; ordered pairs contribute $Z_{\mathcal{B}}^2$, the diagonal contributes $\sum_B N(B)^{-2s} = Z_{\mathcal{B}}(2s)$, and unordered pairs with repetition are half their sum, the same Pólya computation as in Proposition 3.6. (4.4): independence of the occupation numbers gives the Euler product, and $\log \zeta(s) = \sum_p \sum_{k \geq 1} p^{-ks}/k = \sum_k P(ks)/k$. $\square$

Physical reading. Each equation is a named piece of physics. *(4.4) is the free Bose gas*: mode p with occupancy k costs energy $k \log p$, so

$$Z_{\mathcal{M}}(\beta) \;=\; \prod_p \sum_{k \geq 0} e^{-\beta k \log p} \;=\; \prod_p \frac{1}{1 - p^{-\beta}} \;=\; \zeta(\beta):$$

the Euler product *is* the textbook free-boson partition function: the primon gas of Julia [22]. *(4.2) is the distinguishable-particle gas*: ordered k-particle states with no $1/k!$, the counting that produces the Gibbs paradox. *(4.1) is a statistical bootstrap equation* in the sense of Hagedorn–Frautschi [20, 18]: *a fireball is either an input particle or a fusion of two fireballs*, with energy exactly conserved; here the input spectrum is the primons and the fusion is ordered and binary. *(4.3) is the non-planar bootstrap*: unordered fusion, with the $Z(2s)$ term the standard exchange correction for the two branches being *identical*, the same mechanism that separates Bose from Boltzmann counting. Finally, Remark 4.2 places all four in one family.

Remark 4.2 (One gas per statistics). The four right-hand sides are the four classical constructions of the symbolic method [17] (sequence, multiset, planar binary tree, unordered binary tree), each applied to the single-particle element P. Choosing a corner of the square is choosing a *statistics*: what to remember of order and of history. There is one partition function per statistics, all exact functionals of $P(s), P(2s), \ldots$, and the Riemann zeta function is the Bose member of the family. Note where the vacuum lives: only the multiset construction has an empty structure, which is why $n = 1$ belongs to ζ and to no other corner (Remark 2.3).

4.2 The Hagedorn ladder

Theorem 4.3 (Inverse Hagedorn temperatures)**.** *Each series has a finite abscissa of convergence $\sigma_{\mathcal{X}}$, determined by P:*

$$\begin{array}{llll}
\mathcal{M}: & \sigma_{\mathcal{M}} = 1 & \textit{(pole of } \zeta\textit{)}, & Z \to \infty; \\
\mathcal{W}: & P(\sigma_{\mathcal{W}}) = 1, & \sigma_{\mathcal{W}} = 1.3994333287263303\ldots, & Z \to \infty; \\
\mathcal{B}: & 2P(\sigma_{\mathcal{B}}) + Z_{\mathcal{B}}(2\sigma_{\mathcal{B}}) = 1, & \sigma_{\mathcal{B}} = 1.9884768518009081\ldots, & Z \to 1; \\
\mathcal{T}: & P(\sigma^*) = \frac{1}{4}, & \sigma^* = 2.5973851271346716\ldots, & Z \to \frac{1}{2},
\end{array}$$

and $1 < \sigma_{\mathcal{W}} < \sigma_{\mathcal{B}} < \sigma^*$: every remembered layer of structure strictly raises the critical inverse temperature.

Proof sketch. All coefficients are nonnegative, so each abscissa is located by the real singularity. For $\mathcal{W}$, the geometric series diverges exactly when $P \geq 1$. For $\mathcal{T}$, the Catalan series $\sum C_{k-1} P^k$ has radius of convergence $\frac{1}{4}$ in P; since $C_{k-1} 4^{-k} \sim k^{-3/2}/(4\sqrt{\pi})$ is summable, the series still converges at $P = \frac{1}{4}$, with value $\sum_k C_{k-1} 4^{-k} = \frac{1}{2}$. For $\mathcal{B}$, equation (4.3) defines $Z_{\mathcal{B}}$ by descending the argument-doubling cascade ($Z_{\mathcal{B}}(2s)$ lies deep in the convergent region), and the singularity occurs where the discriminant $1 - 2P(s) - Z_{\mathcal{B}}(2s)$ vanishes, at which point $Z_{\mathcal{B}} = 1$: this is Otter's criticality condition for unordered trees [30, 17]. The numerical values are computed to the stated precision in Appendix A; the strict ordering follows from $g_{\mathcal{M}} \leq g_{\mathcal{W}}, g_{\mathcal{B}} \leq g_{\mathcal{T}}$ together with strict inequality on a positive-density set of n. □

Physical reading. The mechanism is Hagedorn's [20]. The number of microstates at energy $E = \log n$ grows like $e^{\sigma_{\mathcal{X}} E}$ (up to powers), so the Boltzmann sum $\sum e^{(\sigma_{\mathcal{X}} - \beta)E}$ diverges for $\beta < \sigma_{\mathcal{X}}$: the gas has a maximal temperature $T_H = 1/\sigma_{\mathcal{X}}$. Remembering order and history multiplies the microstate count at each energy, raises the entropy per unit energy, and lowers the Hagedorn temperature:

gas	states	β_H	$T_H = 1/\beta_H$	transition
Bose (primon)	occupation numbers	1	1	pole
Boltzmann	ordered factorizations	$1.399433\ldots$	$0.714575\ldots$	pole
non-planar fireball	unordered histories	$1.988477\ldots$	$0.502897\ldots$	cusp, $Z_{\mathcal{B}} \to 1$
planar fireball	ordered histories	$2.597385\ldots$	$0.385003\ldots$	cusp, $Z_{\mathcal{T}} \to \frac{1}{2}$

Remark 4.4 (Pole versus cusp: is the Hagedorn temperature attainable?). The symmetrized (associative) gases diverge at their critical point: ζ and $Z_{\mathcal{W}}$ have simple poles ($Z_{\mathcal{W}}$ because $P'(\sigma_{\mathcal{W}}) = -1.7311562\ldots \neq 0$). The history-keeping (non-associative) gases converge at theirs: expanding (4.1) at σ^*,

$$Z_{\mathcal{T}}(s) \;=\; \tfrac{1}{2} - \sqrt{|P'(\sigma^*)|}\,(s-\sigma^*)^{1/2} + O(s-\sigma^*), \qquad \sqrt{|P'(\sigma^*)|} = 0.4796402876\ldots,$$

a square-root branch point with finite value, and similarly $Z_{\mathcal{B}} \to 1$ at $\sigma_{\mathcal{B}}$. The canonical ensemble exists *at* T_H, but the mean energy

$$U(\beta) = -\frac{\partial}{\partial\beta}\log Z_{\mathcal{T}}(\beta) \;\sim\; \frac{\sqrt{|P'(\sigma^*)|}}{2\,Z_{\mathcal{T}}(\sigma^*)}\,(\beta-\sigma^*)^{-1/2} \;\to\; \infty$$

diverges as $\beta \to \sigma^{*+}$: a continuous transition with divergent susceptibility, rather than a divergent partition function. Whether the Hagedorn temperature is attainable, the classic question of the bootstrap literature [29], is decided here by a single algebraic choice: *associativity is a first-order axiom.* The square-root branch point itself is the universal singularity of tree and branched-polymer generating functions (susceptibility exponent $\frac{1}{2}$) [2], the same criticality that governs planar-diagram counting in the Gaussian matrix model, where the Catalan numbers arise as the planar moments and $\frac{1}{4}$ is the edge of the Wigner semicircle [4].

Remark 4.5 (The classical bootstrap on the primon spectrum). The historical statistical bootstrap model fuses in one shot: a fireball is an input particle *or an unordered cluster of* $k \geq 2$ *fireballs*, Boltzmann-weighted by $1/k!$. Its generating equation is $G = \varphi + (e^G - 1 - G)$, whose square-root singularity, first remarked by Nahm [29], occurs when the input function reaches the universal value

$$\varphi_c \;=\; 2\log 2 - 1 \;=\; 0.3862943611\ldots, \qquad G_c = \log 2.$$

Evaluated on the empirical hadron spectrum, this criticality condition is what fixed the original $T_H \approx 160\,\mathrm{MeV}$. Transplanted onto the primon spectrum ($\varphi = P$), it gives the inverse Hagedorn temperatures

$$\beta_H^{\mathrm{Boltz}} = 2.1487511659\ldots, \qquad \beta_H^{\mathrm{Bose}} = 2.3072314551\ldots,$$

the Bose value arising from full multiset statistics ($e^G \mapsto \exp\sum_{k\geq 1} G(ks)/k$, with criticality when $P(s) + 2\sum_{k\geq 2} G(ks)/k$ reaches $2\log 2 - 1$; see Appendix A). These classical gases slot into the same family as the four corners (same input spectrum, same square-root mechanism) but sit off the ladder of Theorem 4.3 because their symmetry-factor weights are fractional rather than integer counts. In bootstrap language, the fireball world $\mathcal{T}$ is a bootstrap model whose decay chains are fully resolved: an integer's prime factorization is its ultimate decay inventory, and the primes are the stable particles.

Remark 4.6 (Growth of the counting functions). By the Wiener–Ikehara theorem applied to $Z_{\mathcal{W}}$ (legitimate: the line $\operatorname{Re} s = \sigma_{\mathcal{W}}$ lies in the half-plane of analyticity of P, and $|P(\sigma_{\mathcal{W}} + it)| < 1$ for $t \neq 0$ by Lemma 5.1 below),

$$\sum_{n \le x} g_{\mathcal{W}}(n) \;\sim\; \frac{x^{\sigma_{\mathcal{W}}}}{\sigma_{\mathcal{W}}\,|P'(\sigma_{\mathcal{W}})|} \;\approx\; 0.4128\,x^{1.39943\ldots}.$$

For the fireball level, transfer heuristics for the square-root singularity suggest $\sum_{n \le x} g_{\mathcal{T}}(n) \asymp x^{\sigma^*}(\log x)^{-3/2}$; making this rigorous is Problem 9.2. Note also the extreme values: along $n = 2^k$ one has $g_{\mathcal{T}}(n) = C_{k-1}$, which grows like $n^2/(\log n)^{3/2}$ up to constants: the microstate count over a single observable can approach the *square* of the observable. For comparison, Kalmár's classical problem [24] of ordered factorizations with arbitrary parts ≥ 2 has Dirichlet series $1/(2 - \zeta(s))$ and growth exponent $\rho = 1.7286472389981836\ldots$ with $\zeta(\rho) = 2$; it sits between our Boltzmann and non-planar rungs but belongs to a different family (its parts need not be stable particles).

5 Fisher zeros of the arithmetic gas

Lee and Yang taught that the thermodynamics of a system is encoded in the complex singularities of its partition function, in complex fugacity for them and in complex temperature for Fisher [26, 15]. We now locate the complex-β singularities of the fireball gas. Since $Z_{\mathcal{T}} = \frac{1}{2}(1 - \sqrt{1 - 4P})$ is an *algebraic* function of P, they come in exactly two kinds: those inherited from the single-particle input P, and the new branch points where the discriminant vanishes.

5.1 What the gas inherits: the Riemann zeros

The prime zeta function continues to the right half-plane by Möbius inversion of (4.4):

$$P(s) \;=\; \sum_{k \ge 1} \frac{\mu(k)}{k} \log \zeta(ks), \qquad \operatorname{Re} s > 0, \tag{5.1}$$

valid off the singular set. Equation (5.1) shows that P has logarithmic branch points at $s = 1/k$ (from the pole of ζ) *and at $s = \rho/k$ for every nontrivial zero ρ of ζ*; these points cluster on the imaginary axis, which is a natural boundary (Landau–Walfisz [25]). Two consequences:

1. Lifting arithmetic to fireballs does not dissolve the Riemann Hypothesis. The zeros of ζ reappear, relocated, inside $Z_{\mathcal{T}}$ as branch points at ρ/k. Where the Hilbert–Pólya program seeks a Hamiltonian whose *spectrum* is the Riemann zeros [3], the arithmetic gas offers the complementary picture: the zeros enter as singularities of the single-particle input, while the gas contributes its own critical singularities on top, and those are computable.

2. No one-variable partition function of this family can contain more analytic information than P itself. Genuinely new invariants must either weight the *shape* of histories (two-variable partition functions, Problem 9.4) or couple to the additive world (§6).

5.2 The new singularities: tree zeros

The new singularities of $Z_{\mathcal{T}}$ are the solutions of

$$P(s) = \tfrac{1}{4}, \tag{5.2}$$

square-root branch points of $Z_{\mathcal{T}}$: the Fisher singularities of the fireball gas. We call them *tree zeros* (they are the zeros of the discriminant $1-4P$). On the real axis there is exactly one, σ^*, since P is strictly decreasing there. Off the axis:

Lemma 5.1. *For $\sigma > 0$ where $P(\sigma)$ converges and $t \neq 0$: $|P(\sigma+it)| < P(\sigma)$.*

Proof. $|\sum_p p^{-\sigma}p^{-it}| \leq \sum_p p^{-\sigma}$ with equality iff all phases $t\log p$ agree modulo 2π. For $t \neq 0$ this fails already for $p = 2, 3$ since $\log 3/\log 2$ is irrational. □

Proposition 5.2. *Every non-real tree zero satisfies* $\operatorname{Re} s < \sigma^*$.

Proof. If $\sigma \geq \sigma^*$ and $t \neq 0$ then $|P(\sigma+it)| < P(\sigma) \leq P(\sigma^*) = \frac{1}{4}$. □

The tree zeros with $0 < t < 50$ are listed in Table 3 (method in Appendix A).

k	$\operatorname{Re} s$	$\operatorname{Im} s$	$t/\frac{2\pi}{\log 2}$
1	1.54427014392	10.0122461654	1.105
2	2.14220467429	17.8224730193	1.966
3	2.24736156780	27.6431435747	3.050
4	2.15502047591	35.5407136013	3.921
5	2.39048394209	45.5394484588	5.024

Table 3: All solutions of $P(s) = \frac{1}{4}$ with $1.4 < \operatorname{Re} s < 2.7$, $0 < \operatorname{Im} s < 50$ (each accompanied by its complex conjugate). A coarser sweep of $1.05 < \operatorname{Re} s < 1.4$ found no further solutions in this height range.

Two empirical observations. First, the ordinates hug integer multiples of $2\pi/\log 2 = 9.0647\ldots$: the partition function is almost periodic in $\operatorname{Im}\beta$, and the recurrence frequency is set by the *lightest primon*, $\varepsilon_2 = \log 2$, whose Boltzmann factor $2^{-\beta}$ dominates P; the interference of $3^{-\beta}$ and later modes then shifts each singularity (violently so for $k = 1$). Second, and decisively: *the real parts wander.* The naive tree analogue of the Riemann Hypothesis (Fisher singularities on a vertical line) is simply false. The correct lifted question is distributional, and it has a precise answer.

5.3 The density of tree zeros

Within the half-plane of almost periodicity the tree zeros have an exact asymptotic density, computable from a random-phase model of the gas.

Theorem 5.3 (Density of the Fisher singularities)**.** *For $\sigma > 1$ let $X_\sigma = \sum_p p^{-\sigma}e^{i\theta_p}$ with (θ_p) independent and uniform on $[0, 2\pi)$, and define the* Jensen function *and the* dominance probabilities

$$\Phi(\sigma) = \mathbb{E}\,\log\big|X_\sigma - \tfrac{1}{4}\big|, \qquad q_p(\sigma) = \mathbb{P}\Big(\,\big|X_\sigma - p^{-\sigma}e^{i\theta_p} - \tfrac{1}{4}\big| < p^{-\sigma}\Big).$$

(i) For every $\sigma > 1$, $\frac{1}{T}\int_0^T \log|P(\sigma+it) - \frac{1}{4}|\,dt \to \Phi(\sigma)$ as $T \to \infty$; Φ is convex, and $\Phi(\sigma) = \log\frac{1}{4}$ for $\sigma \geq \sigma^*$.

(ii) *For $1 < \sigma_1 < \sigma_2$, the number $N_{[\sigma_1,\sigma_2]}(T)$ of tree zeros with real part in $[\sigma_1, \sigma_2]$ and $0 < t \le T$ satisfies*

$$\frac{N_{[\sigma_1,\sigma_2]}(T)}{T} \longrightarrow \frac{\Phi'(\sigma_2) - \Phi'(\sigma_1)}{2\pi}.$$

(iii) $\Phi'(\sigma) = -\sum_p (\log p)\, q_p(\sigma)$. *In particular the density of tree zeros to the right of the line* $\operatorname{Re} s = \sigma_1$ *is* $\frac{1}{2\pi}\sum_p (\log p)\, q_p(\sigma_1)$, *and* $q_p(\sigma) = 0$ *for* $\sigma \ge \sigma^*$, *reproving Proposition 5.2.*

Proof sketch. In closed substrips of $\operatorname{Re} s > 1$ the series $P(s) - \frac{1}{4}$ converges absolutely and uniformly, so it is a uniformly almost periodic analytic function with frequency set $\{\log p\}$, linearly independent over $\mathbb{Q}$ (once more unique factorization). By Kronecker–Weyl the flow $t \mapsto (t \log p \bmod 2\pi)_p$ equidistributes on the torus, identifying time averages with expectations over the phases (θ_p): this is Bohr and Jessen's argument for $\log \zeta$ [10], and gives (i). The value $\log \frac{1}{4}$ for $\sigma \ge \sigma^*$ follows by integrating out the phases one at a time: if $\sum_p p^{-\sigma} \le \frac{1}{4}$ then $|-\frac{1}{4} + \text{rest}| \ge p^{-\sigma}$ pointwise for each peeled mode, and the circular average of $\log |W + ae^{i\theta}|$ equals $\log |W|$ whenever $|W| \ge a$. Statement (ii) is the fundamental theorem of Jessen–Tornehave on mean motions and zeros of almost periodic functions [21]. For (iii), differentiate under the expectation and average over θ_p first: for fixed W with $|W| \ne a$,

$$\frac{1}{2\pi}\int_0^{2\pi} \frac{ae^{i\theta}}{W + ae^{i\theta}}\, d\theta \;=\; \begin{cases} 1 & |W| < a, \\ 0 & |W| > a, \end{cases}$$

by residues. The interchanges are justified: the characteristic function of X_σ is $\prod_p J_0(p^{-\sigma}|\xi|)$, with infinitely many Bessel factors each decaying like $|\xi|^{-1/2}$, so X_σ has a smooth bounded density. Hence $\mathbb{E}\,|X_\sigma - \frac{1}{4}|^{-1}$ is finite, uniformly on compact σ-intervals; this dominates the σ-derivative, whose modulus is at most $\sum_p (\log p)\, p^{-\sigma} / |X_\sigma - \frac{1}{4}|$, and gives the boundary event $|W| = a$ probability zero. ☐

Physical reading. Fix σ and let t run: $P(\sigma + it) - \frac{1}{4}$ is a chain of clock hands, one per prime, the p-hand of length $p^{-\sigma}$ turning at angular speed $\log p$, the whole chain anchored at $-\frac{1}{4}$. A tree zero at height t is the event that the tip of the chain passes through the origin. Whenever one hand is longer than the anchored sum of all the others, that hand drags the tip around the origin and forces one zero per revolution: this is the event measured by q_p. At $\sigma = 1.30$ the hands have lengths $2^{-1.3} = 0.41$, $3^{-1.3} = 0.24$, $5^{-1.3} = 0.12$, …, and the Monte Carlo values are $q_2 = 0.624$, $q_3 = 0.148$, $q_5 = 0.042$, $q_7 = 0.017$ (Appendix A). Weighted by speed, the lightest primon accounts for only 59% of the singularities, which is exactly why the observed density *exceeds* the naive one-per-period guess $\log 2/2\pi$. Chains of rotating arms are the classical mean-motion problem of celestial mechanics, Lagrange's secular perturbations; the Jessen–Tornehave theory invoked above is its solution.

The numbers confirm the theorem. A census to height 500 (Appendix A) finds exactly 59 zeros with $\operatorname{Re} s > 1.3$; the dominance formula predicts $\frac{500}{2\pi}\sum_p (\log p)\, q_p(1.30) = 58.3$, agreement to one percent between two independent computations (Newton root-finding on `primezeta` against a random-phase Monte Carlo). One-per-period would give 55.2. The running counts are $N(100), \ldots, N(500) = 11, 23, 35, 47, 59$; the mean spacing is 8.42 against $2\pi/\log 2 = 9.06$; individual spacings range from 1.31 to 10.54. The left edge, meanwhile, drifts: a coarser sweep of $1.05 < \operatorname{Re} s < 1.30$ finds at least eight further zeros below the strip, with real parts as low as 1.0130 (at $t \approx 470.2$). The real parts approach 1 as the height grows, where almost periodicity fails and the continuation (5.1), with its Riemann-zero branch points, takes over: what happens there is Problem 9.1.

6 The additive bridge: states, processes, and classical observables

The additive world of the theory is flat: $\mathcal{A} = \{\, n\,\ell : n \in \mathbb{N}_{\geq 1}\}$, heaps of a unit quantum ℓ ("one liter"), with $+\colon \mathcal{A} \times \mathcal{A} \to \mathcal{A}$ ordinary accumulation. The question left open so far is how $\mathcal{T}$ and $\mathcal{A}$ interact. The answer proposed here: they do not interact by a map $\mathcal{T} \to \mathcal{A}$. *Multiplication acts on quantities* (fireballs are *processes*, not amounts), and the output of the action is neither a tree nor a flat quantity but a third kind of object: a state of clustered matter.

6.1 States: grouped quantities

Definition 6.1 (Grouped quantities)**.** Define *packages* t and *quantities* f by the mutual grammar

$$t \;::=\; \ell \;\mid\; \boxed{f}, \qquad\qquad f \;::=\; t_1 \oplus t_2 \oplus \cdots \oplus t_m \quad (m \geq 1),$$

where $\boxed{f}$ is a box containing the quantity f, and $\oplus$ is concatenation of finite sequences of packages. Let $\mathcal{G}$ denote the set of quantities. For $k \geq 2$ define the *uniform grouping*

$$k \odot f \;=\; \underbrace{\boxed{f} \oplus \boxed{f} \oplus \cdots \oplus \boxed{f}}_{k}.$$

The *flattening* $\alpha\colon \mathcal{G} \to \mathcal{A}$ counts atoms: $\alpha(\ell) = \ell$, $\alpha(\boxed{f}) = \alpha(f)$, $\alpha(f \oplus g) = \alpha(f) + \alpha(g)$. We identify $\mathcal{A}$ with $(\mathbb{N}_{\geq 1}, +)$ and write $\alpha(f) \in \mathbb{N}_{\geq 1}$.

Physical reading. A quantity is matter with a cluster hierarchy: quanta in bottles, bottles in cartons; nucleons in nuclei, nuclei in atoms. $\oplus$ places systems side by side; α is the extensive observable, total particle number, blind to packaging. $\mathcal{G}$ refines $\mathcal{A}$: a state remembers how it is packed. ($\oplus$ is associative by construction, being concatenation; one may further quotient $\mathcal{G}$ by reordering of packages, and nothing below changes, as the proofs never use the order.)

Definition 6.2 (The action)**.** Define $\rho\colon \mathcal{T} \to \mathrm{Maps}(\mathcal{G}, \mathcal{G})$ by

$$\rho(L_p)(f) = p \odot f, \qquad \rho(T_1 \boxtimes T_2) = \rho(T_1) \circ \rho(T_2).$$

Example 6.3. $\rho(L_3)(\ell) = \boxed{\ell}\boxed{\ell}\boxed{\ell}$: three bottles of one liter. Then

$$\rho(L_2 \boxtimes L_3)(\ell) = 2 \odot \big(\boxed{\ell}\boxed{\ell}\boxed{\ell}\big) = \boxed{\boxed{\ell}\boxed{\ell}\boxed{\ell}}\,\boxed{\boxed{\ell}\boxed{\ell}\boxed{\ell}},$$

two cartons each holding three bottles; whereas $\rho(L_3 \boxtimes L_2)(\ell)$ is three cartons each holding two bottles. Neither is "six liters"; both flatten to 6ℓ. The slogan of §1 is now a statement about the image of ρ: multiplication outputs clustered states, never flat ones.

6.2 The stratification theorems

The three classical laws now separate into three different layers of observability.

Theorem 6.4 (Associativity is additively invisible, exactly)**.** *The action ρ factors through the associativity quotient $\mathcal{T} \to \mathcal{W}$, and the induced map $\mathcal{W} \to \mathrm{Maps}(\mathcal{G}, \mathcal{G})$ is injective. Consequently the congruence $\{(T, T') : \rho(T) = \rho(T')\}$ is precisely $\approx_a$.*

Proof. Factoring: $\mathrm{Maps}(\mathcal{G}, \mathcal{G})$ is a monoid under composition, composition is associative, and ρ sends $\boxtimes$ to $\circ$; hence $\rho((T_1 \boxtimes T_2) \boxtimes T_3) = \rho(T_1) \circ \rho(T_2) \circ \rho(T_3) = \rho(T_1 \boxtimes (T_2 \boxtimes T_3))$, and $\rho(T)$ depends only on $\mathrm{word}(T)$.

Injectivity: let $w = p_1 p_2 \cdots p_k$ and evaluate on the single quantum: $F = \rho(w)(\ell) = p_1 \odot \rho(p_2 \cdots p_k)(\ell)$ consists of exactly p_1 identical top-level packages whose common content is $\rho(p_2 \cdots p_k)(\ell)$. So from F one reads off p_1 (the number of top-level packages) and recurses on the content; the recursion terminates at the boxless quantum ℓ. Distinct words therefore give distinct maps. □

Physical reading. Why does associativity feel inevitable, when nothing in §2 imposes it? Because *processes compose*: any semantics that interprets multiplication as something one *does* (anything of the form $\boxtimes \mapsto \circ$ into an endomorphism monoid) must identify at least $\approx_a$, since composition of maps is associative whether we like it or not. Our semantics identifies *exactly* $\approx_a$: nothing more is lost. Interaction history is unobservable in any operational reading of multiplication; to observe it, one would have to make states out of the processes themselves (currying), which is precisely to leave the additive world. By contrast, nothing forces the process monoid to be commutative, and indeed it is not.

Theorem 6.5 (Commutativity survives until expectation)**.** *For all $T \in \mathcal{T}$ and $f \in \mathcal{G}$:*

$$\alpha\big(\rho(T)(f)\big) \;=\; N(T) \cdot \alpha(f).$$

Hence the induced action on flat quantities $\mathcal{A}$ is multiplication by the shadow, and two trees act identically on $\mathcal{A}$ iff $N(T) = N(T')$, i.e. (by Proposition 3.2) iff $T \approx_{ac} T'$. In particular $\rho(L_2 \boxtimes L_3) \neq \rho(L_3 \boxtimes L_2)$ as maps of $\mathcal{G}$, yet both have the same expectation: every observer reading only particle number sees multiplication by 6*.*

Proof. Induction: $\alpha(p \odot f) = p\,\alpha(f)$ since boxes are α-transparent, and $\alpha(\rho(T_1 \boxtimes T_2)(f)) = N(T_1)\,\alpha(\rho(T_2)(f)) = N(T_1)N(T_2)\,\alpha(f)$. The characterization of equal flat actions is unique factorization. Distinctness on $\mathcal{G}$ is Theorem 6.4 (different words). □

The two observation layers thus recover the two axes of the coarse-graining square: *clustered* observation quotients by exactly associativity; *extensive* observation quotients by exactly associativity and commutativity, and the "exactly" of the second statement is one more disguise of the Fundamental Theorem of Arithmetic.

Theorem 6.6 (Distributivity holds only in expectation)**.** *For every $T \in \mathcal{T}$ and $f, g \in \mathcal{G}$:*

$$\rho(T)(f \oplus g) \;\neq\; \rho(T)(f) \oplus \rho(T)(g) \quad \textit{in } \mathcal{G}, \qquad \textit{yet} \qquad \alpha\big(\rho(T)(f \oplus g)\big) \;=\; \alpha\big(\rho(T)(f) \oplus \rho(T)(g)\big).$$

Proof. Let p_1 be the first letter of $\mathrm{word}(T)$. The left state consists of p_1 top-level packages; the right state consists of $2p_1$; since $p_1 \geq 2$ these differ. Flattening both sides gives $N(T)(\alpha f + \alpha g)$ by Theorem 6.5. □

"Process the merged system" and "process the parts, then merge" are genuinely different protocols with the same extensive outcome: the informal claim of §1, now a theorem. Note also the built-in one-sidedness: there is no $(f \oplus g) \cdot T$ at all, because states are not processes. It is striking that in Loday's arithmetree [27], where both arguments *are* trees, distributivity likewise survives on one side only.

law	in $\mathcal{T}$ (microstates)	on states $\mathcal{G}$	in expectation (α)
associativity	fails (free)	**holds**	holds
commutativity	fails (free)	fails	**holds**
distributivity	not formulable	fails	**holds**

6.3 Where the primes come from

The theory so far imports its particle spectrum from classical arithmetic. The bridge removes this dependency: the primes can be *found* inside $\mathcal{G}$.

Definition 6.7. Call $n \in \mathbb{N}_{\geq 2}$ *groupable* if some quantity consisting of $k \geq 2$ identical packages, each containing $m \geq 2$ quanta, flattens to n, that is, if $n = km$ with $k, m \geq 2$. Call n *prime* if it is not groupable.

Physical reading. A size is composite iff a heap of that size can crystallize into k identical unit cells; the primes are the non-crystallizable sizes, the stable particles of the decay chains of Remark 4.5. This is, of course, the classical definition of primality; that is the point. It is expressible entirely inside the clustered additive world, using only $\odot$ and α, with no imported number labels. One can therefore *define* the particle spectrum as the set of non-groupable heap sizes and generate $\mathcal{T}$ freely on it; the numerical label of a primon is recovered as the observable α of its heap. The foundational question "are prime leaves labeled by known primes, or can the labels be derived?" is thus answered: derived. What cannot be derived from the free multiplicative structure alone (§1.4) becomes derivable the moment the additive world and the action are in place, because primality is an observational property of clustering.

7 Arithmetic functions on trees

Classical arithmetic functions pull back along N. For the Euler totient,

$$\varphi_{\mathcal{T}}(T) \;=\; N(T) \prod_{p \in \operatorname{supp}(T)} \Big(1 - \frac{1}{p}\Big) \;=\; \varphi\big(N(T)\big)$$

by Euler's product formula; e.g. $\varphi_{\mathcal{T}}(L_2 \boxtimes L_3) = 6 \cdot \frac{1}{2} \cdot \frac{2}{3} = 2$. But this pullback factors through the double quotient: it cannot distinguish $L_2 \boxtimes L_3$ from $L_3 \boxtimes L_2$. A genuinely history-sensitive totient should satisfy: (i) a recursion in $\boxtimes$; (ii) compatibility, i.e. it descends to $\varphi \circ N$ after the double quotient; (iii) sensitivity, i.e. it separates some pair of trees with equal shadow. No canonical candidate is known to us; we record this as Problem 9.5. The same design brief applies to Möbius μ, divisor functions, and to two-variable partition functions $Z_{\mathcal{T}}(s, q) = \sum_T q^{\operatorname{stat}(T)} N(T)^{-s}$ for shape statistics stat (depth, right-branch count, Tamari rank): the interesting ones are exactly those *not* expressible through P alone (§5).

8 Computational content

If the primes are the particles of this gas, can the gas help us *find* them? The question deserves a precise answer, because the temptation to answer yes is strong, and wrong in an instructive way. Two theorems mark the boundary: microstate counting is blind to primality (§8.1), while the microstate view describes exactly what a random integer looks like (§8.2). We then identify the one interface where the theory genuinely touches the complexity of primes, the cost of *expressions* (§8.3), and close with a map of what "finding a prime" actually costs and where the open frontier lies (§8.4).

8.1 No free lunch: degeneracy is shape-blind

Proposition 8.1 (No free lunch). *Let $n \geq 2$ have factorization $\prod_p p^{\alpha_p}$.*

(i) $$[\,n \text{ prime}\,] \;=\; g_{\mathcal{T}}(n) \;-\; \sum_{\substack{d \mid n \\ 1<d<n}} g_{\mathcal{T}}(d)\, g_{\mathcal{T}}(n/d).$$

(ii) For every corner $\mathcal{X}$ of the square, $g_{\mathcal{X}}(n)$ depends only on the exponent multiset $\{\alpha_p\}$, not on which primes carry the exponents.

(iii) Given the factorization of n, the values $g_{\mathcal{T}}(n)$ and $g_{\mathcal{W}}(n)$ are computable in time polynomial in $\log n$, and $g_{\mathcal{B}}(n)$ in time $n^{o(1)}$.

Proof. (i) is the defining recursion of $g_{\mathcal{T}}$ rearranged. (ii): for $\mathcal{T}$ and $\mathcal{W}$ by the closed formulas of Propositions 3.4–3.5; for $\mathcal{B}$ by induction along the divisor lattice, which is a product of chains of lengths α_p and hence determined by the exponents alone (the square condition reads "all α_p even"); for $\mathcal{M}$ trivially. (iii): the closed formulas involve factorials of $\Omega(n) = O(\log n)$, hence numbers with $O(\log n \log\log n)$ bits; the $\mathcal{B}$ recursion runs over the divisor lattice, of size $d(n) = n^{o(1)}$. □

Physical reading. The degeneracies cannot even distinguish $12 = 2^2 \cdot 3$ from $18 = 2 \cdot 3^2$ from $20 = 2^2 \cdot 5$ (all have $g_{\mathcal{T}} = 6$, $g_{\mathcal{B}} = 2$, $g_{\mathcal{W}} = 3$): they see the occupation *profile*, never the identities of the occupied modes. Identity (i) does determine primality, but its right-hand side consumes the divisor structure of n, precisely what one would be trying to find, and by (ii)–(iii) the entire structural layer is cheap dressing over the factorization. This is the computational face of the caution of §1.4: freeness means the fireball layer is polynomial-time syntax over its input spectrum. Counting histories, at any corner of the square, can never be the hard step; whatever algorithmic leverage exists must come from outside the counting.

8.2 The random fireball

What the microstate view *does* compute is the statistics of a typical integer seen at full resolution.

Theorem 8.2 (Random fireball). *Draw n uniformly from $\{2, \ldots, \lfloor x \rfloor\}$, then draw T uniformly from the fiber $N^{-1}(n) \subset \mathcal{T}$.*

(i) Conditionally on n, the shape and the leaf word of T are independent, the shape uniform on the $C_{\Omega(n)-1}$ binary shapes and the word uniform on the $\Omega(n)!/\prod_p \alpha_p!$ orderings of the prime multiset.

(ii) As $x \to \infty$, $\big(\Omega(n) - \log\log x\big)/\sqrt{\log\log x}$ converges in distribution to a standard Gaussian.

(iii) Consequently, conditionally on $\Omega(n) = k$ the fireball is exactly a uniform random binary tree with k leaves; in particular its expected height, given k, is asymptotic to $2\sqrt{\pi k}$.

Proof. (i) The bijection of Proposition 3.4 identifies the fiber with (shapes) × (words); the uniform measure on a product set has independent uniform marginals. (ii) is the Erdős–Kac theorem [14] for $\omega(n)$, transferred to Ω: since $\mathbb{E}[\Omega - \omega] = \frac{1}{x}\sum_p \sum_{j \geq 2} \lfloor x/p^j \rfloor \leq \sum_p \frac{1}{p(p-1)} = O(1)$ and $\Omega \geq \omega$, the difference is $o(\sqrt{\log\log x})$ in probability. (iii) The first claim restates (i); the height asymptotic for uniform binary trees is Flajolet–Odlyzko [16]. □

Physical reading. At full resolution a random integer is a small branched polymer: a Gaussian number of constituents concentrated at $\log\log x$ (for $x = 10^{100}$, five or six primons), a shape drawn from the *universal* Catalan ensemble carrying no arithmetic information at all (Proposition 8.1(ii) again), and height of order $\sqrt{\log\log x}$. Erdős–Kac becomes the central limit theorem for the constituent number of the arithmetic gas, and the literature on random Catalan trees (profiles, path lengths, continuum limits) imports wholesale into arithmetic through (iii). Sampling is equally well organized: one can generate a uniform random integer below x *together with its Fock state* in polynomial expected time, using primality tests as an oracle (Bach [7], simplified by Kalai [23]): the occupation-number description is efficiently samplable, even though deterministically *aiming* at a prime is not known to be.

8.3 Expression cost: the invariant that escapes P

Everything so far in this paper is a functional of the input spectrum: the partition functions of §4 through $P(s), P(2s), \ldots$ (§5), the degeneracies through the exponent profile (Proposition 8.1). Here is the first invariant that is not. Extend the grammar of §2 with addition, as anticipated by the bridge: define *expressions*

$$E \ ::= \ 1 \ \mid \ (E_1 \oplus E_2) \ \mid \ (E_1 \boxtimes E_2),$$

with evaluator $\mathrm{ev}(1) = 1$, $\mathrm{ev}(E_1 \oplus E_2) = \mathrm{ev}(E_1) + \mathrm{ev}(E_2)$, $\mathrm{ev}(E_1 \boxtimes E_2) = \mathrm{ev}(E_1)\,\mathrm{ev}(E_2)$, and cost the number of 1-leaves. The minimum cost

$$\|n\| \ = \ \min\{\,\mathrm{cost}(E) : \mathrm{ev}(E) = n\,\}$$

is the classical *integer complexity* of Mahler–Popken [28] (OEIS A005245 [33]), with $3\log_3 n \leq \|n\| \leq 3\log_2 n$ for $n > 1$; allowing subexpression reuse (straight-line programs) gives the Shub–Smale measure $\tau(n) = O(\log n)$ [32].

Cost sees what no degeneracy can: $\|4\| = 4$ while $\|9\| = 6$, although $4 = 2^2$ and $9 = 3^2$ have the same exponent profile. It is, structurally, a bridge quantity: it measures how expensively a fireball can be synthesized once additions are allowed inside the build. And it is exactly where the known connections between the structure of multiplication and the complexity of primes live:

Proposition 8.3 (Wilson's theorem as an expression statement). *(i) $n \geq 2$ is prime if and only if $(n-1)! \equiv -1 \pmod{n}$.*

(ii) If $(x, N) \mapsto x! \bmod N$ were computable in time polynomial in $\log N$, then factoring would be in polynomial time.

Proof. (i) is Wilson's theorem. (ii): $\gcd(x! \bmod N, N) = \gcd(x!, N)$, which equals 1 exactly when x is smaller than the least prime factor of N and exceeds 1 from that point on; binary search over $x \in [1, N]$ therefore finds the least prime factor with $O(\log N)$ evaluations, and recursion factors N completely. □

The same object carries the deeper conjectures. If $n!$ had straight-line programs of size polylogarithmic in n, factoring would have polynomial-size circuits (folklore; see Blum–Cucker–Shub–Smale [9]), and the Shub–Smale τ-conjecture (a polynomial bound on the integer roots of a polynomial in terms of its program size) implies $P_{\mathbb{C}} \neq NP_{\mathbb{C}}$ [32]. In the vocabulary of this paper: *the computational hardness of arithmetic is a statement about the additive–multiplicative interface of §6, about the cost of synthesizing specific fireballs with additions, and Wilson's theorem says primality is one factorial-expression away.*

The theory also suggests its own cost observable. Define the *factored cost* and the *addition dividend*

$$A(n) \;=\; \sum_{p^a \| n} a\,\|p\|, \qquad \delta(n) \;=\; A(n) - \|n\| \;\geq\; 0:$$

$A(n)$ is the cost of the cheapest expression that respects the prime factorization (synthesize each primon additively, then fuse), and $\delta(n)$ is the saving that additions *above* the factorization can buy. The long-standing open question whether $\|2^k\| = 2k$ is exactly the statement $\delta \equiv 0$ on powers of two; and δ is not identically zero: $\delta(55) \geq 1$, since $A(55) = \|5\| + \|11\| = 13$ while $55 = 2 \cdot 27 + 1$ gives $\|55\| \leq 12$. Nothing about the distribution of δ appears to be known; see Problem 9.7.

8.4 What finding a prime costs

Finally, the target itself, stated honestly. *Recognizing* primes is solved: PRIMES $\in$ P [1]. *Hitting* one randomly is easy: by the prime number theorem a random n-bit integer is prime with probability $\sim 1/(n \log 2)$, so sampling plus testing generates an n-bit prime in expected time $\widetilde{O}(n^3)$ [13]. The genuine open problem, the Polymath 4 problem [31], is *deterministic* generation:

task	best known	status
test an n-bit integer	deterministic poly(n) [1]	solved
generate, randomized	$\widetilde{O}(n^3)$: sample and test	routine
generate, deterministic	$\sim 2^{n/2}$: analytic $\pi(x)$ methods [31]	open, even under RH
generate, pseudodeterministic	poly(n), infinitely many n [11]	frontier
quantum search	$\sim 2^{0.2625\,n}$: Grover [19] on [8]	quadratic-type gain

Even the Riemann Hypothesis only bounds prime gaps by $O(\sqrt{x}\log x)$, leaving a $2^{n/2}$ search; a Cramér-type gap bound $O(\log^2 x)$ [12] would make "scan forward and test" polynomial. The two attack surfaces are exactly the two places this paper has located all remaining hardness. *Gaps* are quantitative statements about the additive–multiplicative interface: §6 names the interface but is not yet quantitative. *Derandomization*: the primes are dense and efficiently recognizable, so deterministic generation is a problem of *aiming* (pseudorandomness rather than number-theoretic search), and the natural formal home of "dense yet hard to aim at" is expression cost (§8.3): a deterministic generator is, in the end, a short expression that evaluates to a prime. The honest summary: the theory contributes a sharp map of where the hardness cannot be (Proposition 8.1), well-posed cost questions at the interface where it can be, and no shortcut: by its own no-free-lunch proposition, none was available from counting alone.

9 Open problems

Question 9.1 (Fisher-zero distribution: beyond the almost periodic region)**.** Theorem 5.3 settles the density of tree zeros in substrips of $\operatorname{Re} s > 1$. Remaining: (a) the left edge: the census finds real parts as low as 1.013 by height 500; do they accumulate at 1? Do zeros of the continued $P - \frac{1}{4}$ exist with $\operatorname{Re} s \leq 1$, where almost periodicity fails and the Riemann zeros enter through (5.1)? (b) the total density $\lim_{\sigma_1 \downarrow 1} \frac{1}{2\pi} \sum_p (\log p)\, q_p(\sigma_1)$: expected finite (the tail should behave like $\sum_p (\log p)\, p^{-2\sigma}$), but its value and the speed of the leftward drift are open. (c) pair correlations and the gap law of the ordinates: the census gives spacings from 1.31 to 10.54 around mean 8.42; is there level repulsion? (d) a Lee–Yang-type question: is there a natural deformation of the gas for which the singularities align on a line? (e) the same program for the level sets $P(s) = c$, $0 < c < 1$: the tree zeros are $c = \frac{1}{4}$, the Boltzmann-gas poles $c = 1$.

Question 9.2 (Tauberian rigor)**.** Prove $\sum_{n \leq x} g_{\mathcal{T}}(n) \sim c\, x^{\sigma^*} (\log x)^{-3/2}$ (a Delange-type theorem for Dirichlet series with a square-root singularity), and its analogue for $g_{\mathcal{B}}$.

Question 9.3 (The cascade of the non-planar gas)**.** The continuation

$$Z_{\mathcal{B}}(s) = 1 - \sqrt{1 - 2P(s) - Z_{\mathcal{B}}(2s)}$$

propagates each singularity of $Z_{\mathcal{B}}$ at s_0 to potential singularities at $s_0/2, s_0/4, \ldots$ and mixes them with those of P. Map this self-similar singularity structure; determine whether $Z_{\mathcal{B}}$ has a natural boundary strictly to the right of $\operatorname{Re} s = 0$.

Question 9.4 (Beyond the single-particle input)**.** Every one-variable partition function of §4 is a functional of $P(s), P(2s), \ldots$ Find a natural two-variable $Z_{\mathcal{T}}(s, q)$ (weighting a shape statistic of the history) that is provably *not* such a functional, and study its phase diagram in (s, q).

Question 9.5 (History-sensitive totient)**.** Construct φ^{tree} satisfying (i)–(iii) of §7, ideally with an inclusion–exclusion (Möbius) interpretation over subtrees.

Question 9.6 (The full expression calculus)**.** §6 lets processes act on states, and §8.3 equips the two-sorted expression grammar with a cost. What is missing is its *rewriting theory*: treat distributivity as a directed rewrite rule (an evaluation step) rather than an equation, study normal forms and confluence, and relate them to Loday's grove addition [27, 6].

Question 9.7 (The addition dividend)**.** Determine the behavior of $\delta(n) = A(n) - \|n\|$ (§8.3). Is $\delta \equiv 0$ on powers of two (equivalently, $\|2^k\| = 2k$)? Is δ unbounded? What is its average order? Is $\|n\|$ computable in time polynomial in $\log n$? And the structured refinement: quantify the *cost of remembering*, the minimal cost of an expression whose multiplicative skeleton is constrained to a fixed corner of the coarse-graining square.

Question 9.8 (Axiomatics of the bridge)**.** Characterize the pair $(\mathcal{G}, \alpha)$ abstractly: for which "clustering worlds" does Theorem 6.4 hold with *exactly* $\approx_a$, and is $(\mathcal{G}, \alpha)$ universal among them?

A Computational verification

All computations are reproducible from the accompanying `primeleaf` Python package (see the repository `README` for installation): `python -m primeleaf verify-combinatorics` reproduces the combinatorial checks below, `verify-zeta` the identities and constants, and `census` and `dominance` the zero census and the density check.

A.1 Combinatorial checks

For every $n \leq 20000$: the closed formula of Proposition 3.4 agrees with a direct recursive enumeration $g_{\mathcal{T}}(n) = [n \in \mathbb{P}] + \sum_{d|n,\, 1<d<n} g_{\mathcal{T}}(d)\, g_{\mathcal{T}}(n/d)$; the multinomial of Proposition 3.5 agrees with the first-letter recursion; and the unordered recursion of Proposition 3.6 reproduces the Wedderburn–Etherington numbers $1, 1, 1, 2, 3, 6, 11, 23, 46, 98, 207, 451, 983, 2179$ at $n = 2^k$, $k \leq 14$, as well as the hand-enumerated values $g_{\mathcal{B}}(12) = 2$, $g_{\mathcal{B}}(24) = 4$, $g_{\mathcal{B}}(30) = 3$. The OEIS near-miss A375120 (which uses ordered pairs on the diagonal) first diverges at $n = 144$: 42 against the true 41; a control query confirmed the OEIS search interface finds A375120 from its own terms while the $g_{\mathcal{B}}$ terms return no match (July 2026).

A.2 Analytic checks and constants

With $g_{\mathcal{T}}(n)$, $g_{\mathcal{B}}(n)$ enumerated for $n \leq 20000$, partial sums of the Dirichlet series were compared with the closed/functional forms of Theorem 4.1:

series	s	partial sum vs. closed form	difference
$Z_{\mathcal{T}}$	3	$0.225547637865\ldots$ vs. $0.225705704214\ldots$	$-1.6 \cdot 10^{-4}$ (tail)
$Z_{\mathcal{T}}$	4	$0.084059063652\ldots$ vs. $0.084059066410\ldots$	$-2.8 \cdot 10^{-9}$
$Z_{\mathcal{T}}$	6	agreement	$-3.1 \cdot 10^{-18}$
$Z_{\mathcal{B}}$	3	agreement	$-4.0 \cdot 10^{-7}$
$Z_{\mathcal{B}}$	4	agreement	$-1.1 \cdot 10^{-11}$

the residual differences being exactly the truncation tails expected from the critical exponents. Constants (via `mpmath.primezeta` at 25–30 significant digits):

$$\begin{aligned} \sigma^* &= 2.59738512713467162\ldots, & P'(\sigma^*) &= -0.230054805474\ldots, \\ \sigma_{\mathcal{B}} &= 1.98847685180090810\ldots, & Z_{\mathcal{B}}(\sigma_{\mathcal{B}}) &= 1 \text{ (verified to } 10^{-6}), \\ \sigma_{\mathcal{W}} &= 1.39943332872633032\ldots, & P'(\sigma_{\mathcal{W}}) &= -1.73115620186\ldots, \\ \rho_{\text{Kalmár}} &= 1.72864723899818362\ldots, & P(2) &= 0.45224742004106550\ldots \end{aligned}$$

For the classical bootstrap of Remark 4.5 on the primon spectrum: Boltzmann counting, $P(\beta_H) = 2\log 2 - 1$, gives $\beta_H = 2.14875116590273\ldots$; Bose counting, solving $P(s) + 2\sum_{k\geq 2} G(ks)/k = 2\log 2 - 1$ with G computed by the argument-doubling cascade, gives $\beta_H = 2.30723145510135\ldots$ with $G(\beta_H) = 0.6646652572\ldots$

A.3 Tree zeros

The zeros of Table 3 were located by a grid scan of $|P_{\leq 20000}(s) - \frac{1}{4}|$ (truncated prime sum, step 0.02) over $[1.40, 2.70] \times [0.3, 50]$, followed by Newton polishing on the full `primezeta` at 20 significant digits; each reported zero satisfies $|P(s) - \frac{1}{4}| < 10^{-15}$. A coarser sweep of $[1.05, 1.40) \times [0.5, 50]$ (step 0.05×0.25) found no additional candidates; its only sub-threshold dip flowed to zero #1 of the table.

A.4 Zero census and the density check

The census of §5.3 (command `python -m primeleaf census`) extends the scan to $[1.30, 2.72] \times [0.3, 500]$ with primes up to 10^5 (grid 0.02×0.05, seed threshold 0.06, Newton polish at 18 significant digits, residuals below 10^{-14}): 59 zeros, real parts in (1.3149, 2.5396), listed in `data/zeros_t500.txt`. The running counts are $N(100), \ldots, N(500) = 11, 23, 35, 47, 59$; the one-per-period line $T \log 2/2\pi$ would give $11.0, 22.1, 33.1, 44.1, 55.2$; the mean spacing is 8.417, with extremes 1.31 and 10.54. A coarse sweep of $[1.05, 1.30] \times [0.5, 500]$ (steps 0.05×0.25, not exhaustive) finds eight further zeros with real parts down to 1.01298 (at $t \approx 470.24$). The dominance probabilities at $\sigma = 1.30$ were estimated from $4 \cdot 10^5$ random-phase samples over the primes up to 10^4:

$$q_2 = 0.6237, \quad q_3 = 0.1484, \quad q_5 = 0.0416, \quad q_7 = 0.0167, \quad q_{11} = 0.0051, \quad q_{13} = 0.0031,$$

giving predicted density 0.11659 per unit height, i.e. 58.3 zeros to height 500, against 59 observed.

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